

Reading the Evidence for Triangle Congruence

Answer Key

High-school geometry

Quick Answers

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|---|---|
| 1. — | 5. (b) the remaining leg = 12 cm, — |
| 2. 7 | 6. — |
| 3. — | 7. (a) $x = 8$, (b) the corresponding side = 20 cm |
| 4. (a) $x = 10$, (b) the measure of angle A = 46 degrees | 8. — |
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What is verified

3 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 3, 5, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: High-school geometry

HSG-CO.B – *problems 1, 2, 3, 4, 5, 6, 7*

HSG-CO.C – *problem 8*

Difficulty 1–5 of 5 across 14 tagged checks.

- 1.** Three pairs of sides tick-marked. Name the criterion and its three statements.

Criterion: SSS. All three pairs of corresponding sides are marked congruent, and nothing about angles is needed.

The three statements: $\overline{AB} \cong \overline{DE}$ (single ticks), $\overline{BC} \cong \overline{EF}$ (double ticks), $\overline{CA} \cong \overline{FD}$ (triple ticks). Therefore $\triangle ABC \cong \triangle DEF$ by SSS.

Grading: the criterion plus all three pairs, written in corresponding order. Listing the pairs out of correspondence (for example $\overline{AB} \cong \overline{EF}$) is the error to look for.

- 2.** SAS with $AB = 3x + 4$ and $DE = 5x - 10$ cm. Find x .

Solution: congruent triangles have congruent corresponding sides, and $\overline{AB}$ corresponds to $\overline{DE}$, so the two lengths are equal.

$$3x + 4 = 5x - 10$$

$$14 = 2x$$

Check: $3(7) + 4 = 25$ and $5(7) - 10 = 25$.

$$x = 7$$

3. Two pairs of sides marked. (a) What completes SAS? (b) What completes SSS?

(a) A pair of congruent **angles**, and it must be the *included* angle — the angle formed where the two already-marked sides meet, so $\angle B \cong \angle E$ for the marked pair here. Any other angle gives side-side-angle, which is not a criterion: the third side is then free to swing into two positions.

(b) The **third pair of sides** marked congruent. No angle information is needed at all; SSS uses sides only.

Grading: (a) must say *included* (or identify the vertex between the two marked sides). Naming any angle without that condition is the common error.

4. ASA with $m\angle A = 4x + 6$ and $m\angle D = 6x - 14$. (a) x . (b) $m\angle A$.

(a) Corresponding angles of congruent triangles are congruent, so their measures are equal.

$$4x + 6 = 6x - 14$$

$$20 = 2x$$

So $x = 10$.

$$x = 10$$

(b) $m\angle A = 4(10) + 6 = 40 + 6$. Check: $m\angle D = 6(10) - 14 = 46$ too.

$$46^\circ$$

5. Right triangles, hypotenuse 15 and leg 9 cm each. (a) Criterion and why SSA fails otherwise. (b) The remaining leg.

(a) **HL**. Both triangles have a right angle (at C and at Z), and the matching parts are a hypotenuse and a leg — exactly the hypotenuse-leg criterion, which applies to right triangles only.

Why the general pattern fails: two sides and an angle that is *not* between them do not fix a triangle. Swinging the second side from the end of the first gives two different closing positions, so two non-congruent triangles can carry the same three measurements. In a right triangle that second position is unavailable — the leg must meet the other leg at 90° — which is what rescues HL.

(b) By the Pythagorean theorem, the remaining leg is $\sqrt{15^2 - 9^2} = \sqrt{225 - 81} = \sqrt{144}$.

$$BC = 12 \text{ cm}$$

6. Quadrilateral $ABCD$ split by diagonal $\overline{AC}$; $\overline{AB} \cong \overline{AD}$, $\angle BAC \cong \angle DAC$. (a) The shared-side property. (b) The criterion.

(a) The **reflexive property of congruence**: any segment is congruent to itself, so $\overline{AC} \cong \overline{AC}$. It is a legitimate line because both triangles literally contain the same segment — there is nothing to prove, and it is the standard way a shared side enters a proof. No tick mark is needed or expected.

(b) The marked side $\overline{AB} \cong \overline{AD}$, the marked angle $\angle BAC \cong \angle DAC$ *between* them and the shared side, and $\overline{AC} \cong \overline{AC}$ give **SAS**, so $\triangle ABC \cong \triangle ADC$.

Grading: (a) must name the reflexive property, not merely observe that the side is shared.

7. SSS with $AB = 2x + 1$, $DE = x + 9$, $BC = 3x - 4$ cm. (a) x . (b) EF .
(a) $\overline{AB}$ corresponds to $\overline{DE}$:

$$\begin{aligned}2x + 1 &= x + 9 \\x &= 8\end{aligned}$$

Check: $2(8) + 1 = 17$ and $8 + 9 = 17$.

- (b) $\overline{BC}$ corresponds to $\overline{EF}$, so $EF = BC = 3(8) - 4 = 24 - 4$.

$x = 8$

$EF = 20$ cm

8. $\overline{PR}$ and $\overline{QS}$ meet at X ; $\overline{PQ} \cong \overline{SR}$, $\angle QPX \cong \angle RSX$. (a) The hidden pair and the criterion. (b) Why the SSA variant fails.

(a) The hidden pair is $\angle PXQ$ and $\angle SXR$, the two angles at the crossing point. They are **vertical angles**, so they are congruent by the *Vertical Angles Theorem* — no tick mark can show this, it has to be read off the figure.

Now list the three parts in order around each triangle: $\angle QPX \cong \angle RSX$, $\angle PXQ \cong \angle SXR$, and the side $\overline{PQ} \cong \overline{SR}$. That side is *opposite* one of the two angles rather than between them, so the criterion is **AAS**, and $\triangle PQX \cong \triangle SRX$.

(b) The classmate's version keeps $\overline{PQ} \cong \overline{SR}$ and $\angle QPX \cong \angle RSX$ and adds $\overline{QX} \cong \overline{RX}$: two sides and an angle that is not between them. That pattern does not fix a triangle — from the end of the first side the second can swing to two different closing points, so two triangles with those three measurements need not be congruent. Side-side-angle is not a criterion, and the vertical-angle pair the figure actually provides is what makes the real argument work.

Grading: (b) must give the reason (two positions are possible), not merely assert that SSA “is not allowed”.